

Practice Quiz 09B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 09

Well-definedness on a quotient; onto maps and inverses.

Write $[x]_7$ for the congruence class of x modulo 7.

1. Consider

$$f : \mathbf{Z}_7 \longrightarrow \mathbf{Z}_7, \quad f([x]_7) = [3x + 2]_7.$$

(a) Prove that f is well defined.

Solution Suppose $[x]_7 = [y]_7$. Then $7 \mid (x - y)$, so $x - y = 7k$ for some integer k . Therefore

$$(3x + 2) - (3y + 2) = 3(x - y) = 7(3k).$$

Thus $7 \mid ((3x + 2) - (3y + 2))$, so $[3x + 2]_7 = [3y + 2]_7$. Hence the output does not depend on the representative and f is well defined. ■

(b) Prove that f is onto and determine f^{-1} .

Solution The number 5 is an inverse of 3 modulo 7, since $3 \cdot 5 \equiv 1 \pmod{7}$. Given an arbitrary $[y]_7$, choose

$$[x]_7 = [5(y - 2)]_7.$$

Then

$$f([x]_7) = [3 \cdot 5(y - 2) + 2]_7 = [y]_7,$$

because $3 \cdot 5 \equiv 1 \pmod{7}$. Thus f is onto. The calculation also gives

$$f^{-1}([y]_7) = [5(y - 2)]_7.$$

The same representative check as in part (a), with coefficient 5, shows that this inverse rule is well defined. ■